5.5) Grade compensation

- CG formula

CG: compensated grade

$= (G - GC)$ %.

Q Gradient on a highway is 1 in 15. Radius of curve is 150 metre. ~~after~~ findout grade compensation & compensated grade.

Ans: G = 1 in 15

R = 150 M.

$G = \frac{1}{15} \times 100 = 6.67\%$

[check it is more than 4% hence GC & CG are provided]

$GC = \frac{30 + R}{R} = \frac{180}{150} = 1.2\%$

$= \frac{75}{150} = 0.5\%$

GC = 0.5%.

CG = G - GC = 6.67 - 0.5 = 6.17.

Creeperlane:

- kya hota hai

Creeperlane: When the uphill climb is extremely long, it may be desirable to introduce an additional lane so as to allow slow ascending vehicles to be removed from main stream so that fast moving vehicles are not affected.

frontage road :- defn.

Frontage road: it is provided to give access to the properties along an important highway with controlled access as expressway or freeway.

Minimum width of shoulder kya hogi

Minimum width of shoulder recommended by IRC : 2.5m.

Guard rail kya hota kab detc h?

Guard rail is provided when embankment height is more than 3 m.

Q1 Worksheet 1 ka qn 1.
Speed of overtaking & overtaken vehicle are 100 & 50kmph resp....
avg acc 0.92 m/s^2
find OSD
desirable length of ot zone

Ans 1: OSD: $d_1 + d_2 + d_3$

Numericals: $0.278 V_b \times 2 + 2s + b + 0.278 V_c T$.

V_b : 50 kmph

$d_1 = 0.278 \times 50 \times 2 = 27.8$ m.

$T = \sqrt{\frac{4s}{a}}$ $s = 0.7V_b + 2$

$s = 0.7 \times 50 + 6.1 = 41.1$ m

$= \sqrt{\frac{4 \times 41.1}{0.92}} = 13.36$ second

$2s = 2 \times 41.1 = 82.2$ m.

$b = .278 V_b T = 0.278 \times 50 \times 13.36$

$b = 185.7$ m

$d_2 = 2sf + b$

$= 82.2 + 185.7$

$d_2 = 267.8\ m.$

$d_3 = 0.278\ V_c\ T$

→ Value.

$d_3 = 0.278 \times 100 \times 13.36 = 371.4\ m.$

$d = d_1 + d_2 + d_3 = 185.7 + 267.8 + 371.4$

$= 666\ m.$

Desirable length $\int_{0}^{5x} SD = 666$

$= 3330\ m$

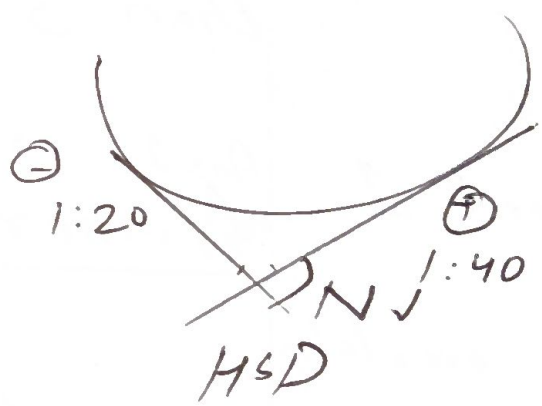
Ans

A. Worksheet 1 ka Q. 5.

A falling gradient of 1 in 20 meets rising gradient of 1 in 40 on main in plain country. S = 120. find length of valley curve which should provide safe driving distance.

(i) $L \geq HSD$

$L = \frac{NS^2}{1.5 + 0.34S}$



given parameters to obtain
S = 120 m (assume).

$N_1 = 1 \text{ in } 20 = \frac{-1}{20} = -0.05$

$N_2 = +\frac{1}{40} = +0.025$

N is difference or deflection L

$= |N_1 - N_2| = |-0.05 - 0.025|$

$= +|-0.075| = 0.075$

$$L : \frac{NS^2}{1.5 + 0.034S} = \frac{0.075 \times 120^2}{1.5 + 0.034 \times 120}$$

$$= 193.5\ m.$$

Check: $0L > S$

$193.5 > 120$ Pass.

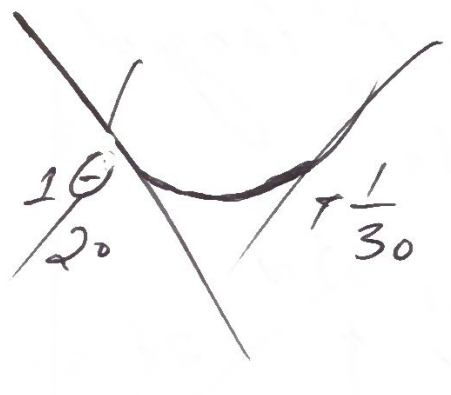
Ans is 193.5 Mk.

for the same question or above qn if $S = 220\ m$ then what will be length of valley curve.

---

8 A valley curve is found or formed by a descending gradient $\frac{1}{20}$ meets with an ascending gradient $\frac{1}{30}$. The design speed of vehicle is 80 kmph the allowable rate of change of centrifugal acceleration is taken as $0.61\ ms^{-3}$. find out length of valley curve for comfort condition.

Ans:



$$N = |N_1 - N_2|$$

$$N_1 = \frac{-1}{20} = -0.05$$

$$N_2 = +\frac{1}{30} = +0.033$$

$$N = |-0.05 - 0.033| = 0.083$$

$$C = 0.61\ ms^{-3}$$

$$V = 80\ kmph.$$

$$L = 0.38\sqrt{NV^3}$$

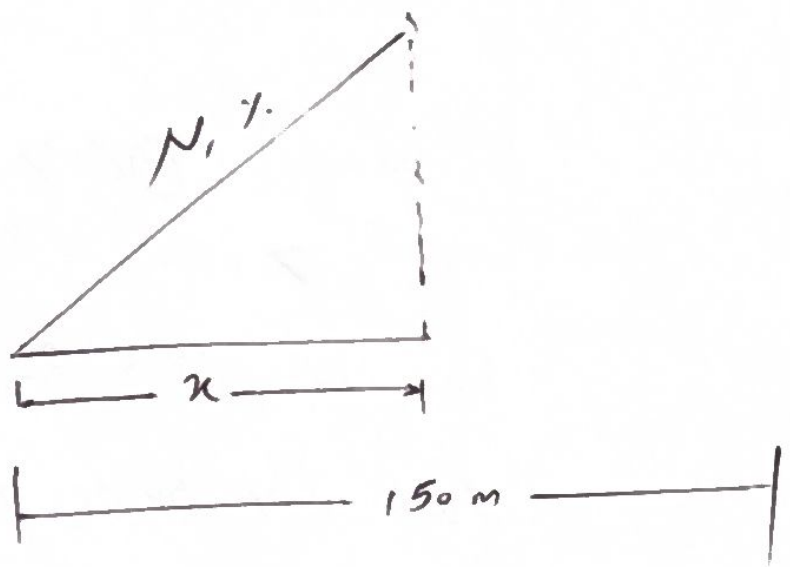
$$= 0.38\sqrt{0.083 \times 80^2}$$

$$= 78.33\ m \quad \text{Ans.}$$

Class 6, 9th Aug 2024

E.1] A rising gradient of a parabolic curve is 4% & falling grade is 2%. Then the highest point on the summit curve is at a distance measured horizontally from 1st tangent will be.

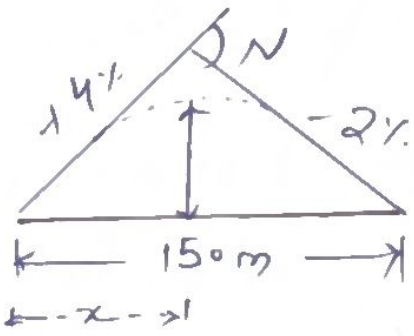
Ans: Parabolic, summit curve



$N = |N_1 - N_2|$

$= |4 - (-2)|$

$= 6\%$ or 0.06



$\frac{N\%}{L} : \frac{N_1\%}{x} \Rightarrow x : \frac{N_1\% \cdot L}{N}$

$: \frac{(4/100) \times 150}{(6/100)} = 100\ m$

Gate ka qn. Very imp.

Q2 A rising gradient 1 in 50 meets a falling gradient 1 in 100 the length of vertical curve if rate of changed gradient is 1% per 100m length is.

$N_1 = 1:50$

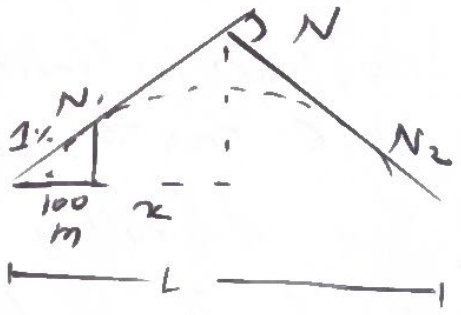
$= 2\%$

$N_2 = -1:100$

1% per 100m

$L = ?$

$N = \left| \frac{1}{50} - \left(-\frac{1}{100}\right) \right| = 0.03 = 3\%$



$$\frac{L}{N} : \frac{x}{N_1} : \frac{100m}{1\%}$$

$$\frac{L}{0.03} : \frac{100}{0.01} : L : 300m$$

alternate method

1% percent $\rightarrow$ distance $\rightarrow$ 100m

N: 3% per distance, $\frac{3\%}{1\%} \times 100$

$= 300m$

Q Calculate SSD for V = 50 kmph for

(1) 2 way traffic for 2 lane road

(2) 2 way traffic in single lane road

Note: Consider values as per IRC

Ans: $SSD = 0.278 V_0 t_R + \frac{V^2}{254(f \pm n\%)}$

$f = 0.35$

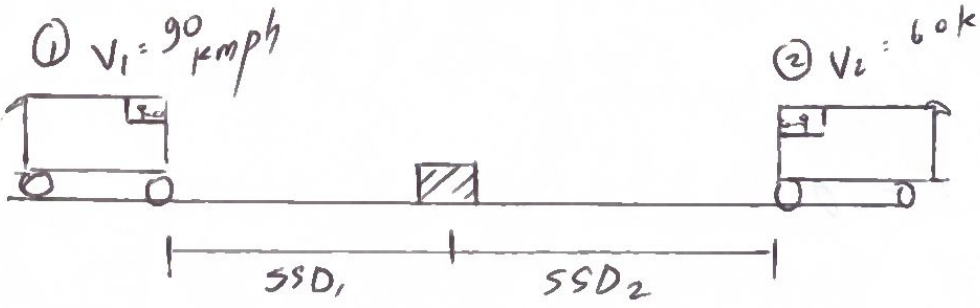
$t_R = 2.5$

$= 0.278 \times 50 \times 2.5 + \frac{50^2}{254 \times (0.35)}$

$= 61.4 \approx 62.8 m$

For (1) SSD: 62.8m for (2) SSD: 255D = 125m

Q What is the minimum sight distance to avoid head on collision of two cars approaching @ 90 kmph and 60 kmph speed use reaction time 2.5 second brake $f = 0.35$ in either case. $f = 0.7$ Coeff of longitudnal friction



$$SSD = V t_R + \frac{V^2}{2g(f \pm N\%)}$$

$$SSD_1 = 0.278 V t_R + \frac{V^2}{2g(f \pm N\%)}$$

90 — 2.5 — 0

SSD_1 : $0.278 \times 90 \times 2.5$

$+ \frac{V^2}{254(f \pm n)}$ → brake efficiency 50%

$0.278 \times 90 \times 2.5 + \frac{90^2}{254 \times 0.7 \times 0.5}$

$= 153.6\ m$

SSD_2 : $0.278 \times 60 \times 2.5 = 82.19\ m$

$+ \frac{60^2}{254 \times 0.7 \times 0.5}$

$SSD = SSD_1 + SSD_2 = 82 + 153$

$= 235.8\ m.$

Q find SSD for descending gradient of 2% for a design speed 80 kmph velocity given $22.22\ m/s^2$ coefficient of longitudinal friction = 0.35

Ans $SSD = vt_R + \frac{v^2}{2g(f \pm N)}$

$SSD = 22.22 \times 2.5 + \frac{22.22^2}{2 \times 9.81 \left[(0.35 \times 1) - \frac{2}{100}\right]}$

$SSD = 131.9\ m.$

(a) What is ISD

Ans @ $ISD = 2\,SSD$

$= 131.9 \times 2$

$= 264\ m$

Traffic engineering

traffic study

- Basic objective
  - ko
  - fac
  - pvmt
  - acc
- units
  - traffic vol
  - traffic s
  - traffic speed
  - traffic cap

Part 2 Traffic engineering & pavement design. / Highway part 2

Traffic engineering (chapter 1)

Traffic study

The basic objective of traffic study

- to obtain knowledge of type and volume of traffic at present, in future
- To study road facility & improvements to be done
- to design geometric features & pavement thickness
- to analyse accidents

(1) Traffic volume (vehicle per hour)

(2) Traffic density (vehicle per km)

(3) traffic speed (kmph)

(4) traffic capacity = traffic volume / lane

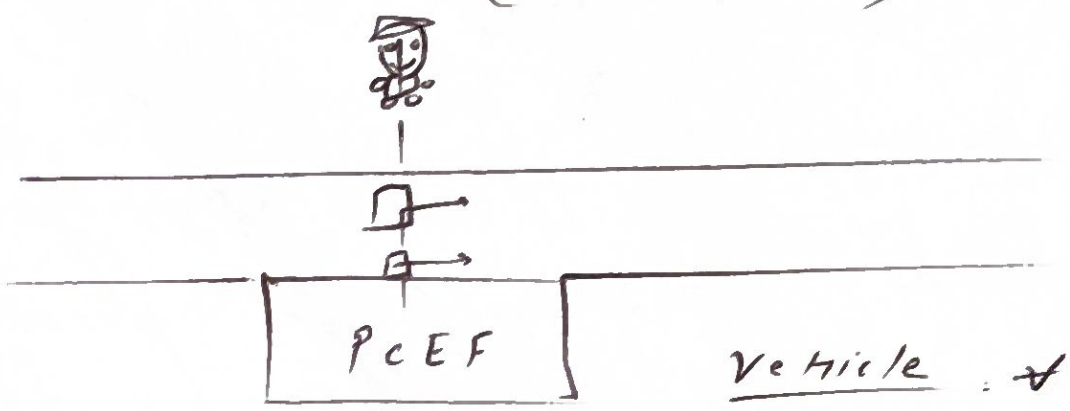
= vehicle/hr / lane

= vehicle/hr/lane

6.5) traffic vol
+ alternate name
— expressed as
— unit
— study
— counting mthds
— man
— mech
— alt name
— intersc.
— mech counting tools
— PN
— PZ
— VC
— bend
— MM

traffic volume / traffic flow

is expressed as the number of vehicle crossing a section of road per unit time. (vehicle/hour)



Vehicle / hour

7:00am – 8:00am

Traffic volume study

1) Counting methods
   1. manual traffic counting
   2. mechanical counting / automatic counted
   3. traffic volume counted @ intersecns

Mechanical counting is done by -

1) Pneumatic tube
2) piezoelectric cable
3) Videocamera
4) Bending plate.
5) Micro mm wave radar detectors

Traffic volume Study
- Present of data
  - ① AADT
    - kyah?
  - ② ADT
    - kya hz.
  - ③ Trend chart
    - kyah?
  - ④ Traffic flow map
    - alternate names
    - use?

Presentation of traffic vol data

① AADT (Average annual daily traffic volume)

The average 24 hr. traffic volume at a given location over full 365 days of year is called AADT.

② ADT (average daily traffic volume)

It is the twenty four hr average traffic volume @ given location for some period of time less than 1 year.

③ Trend chart

Showing traffic volume trend over perid of years.

④ Traffic flow map or diagram along routes

. Generally used to represent traffic @, @ intersections

6.2 Traffic study

- Present mthds
  - ⑤ ★ imp 30th highest hourly vol
    - kya hai?
      - kya design krte h isse
      - congestion okkad hota hai?

⑤ 30th highest hourly vol

The hourly volume that will be exceeded only 29 times in a year & all other hourly volumes of the year will be less than this volume

① Highway facilities are generally for this volume.

② Congestion only during 29th hour in the year.

- ⑥ Peak hr factor
  - kya hai?
  - kis chiz ka measure h?
  - formula

⑥ Peak hour factor PHF

. The hourly volume during the maximum volume hour of the day divided by 15 minute flow rate within the peak hour is called peak hour factor.

. It is a measure of traffic demand fluctuations within peak hour.

. PHF : $\frac{1}{4}$ $\frac{q}{q_{15}}$ or $\frac{1}{4}$ th of { dinka wo hour jisme traffic max / Us hour ke max 15 min ka traffic

Presentation TVS

PHF

- PHF for highway

a The resultant of traffic count taken between 5 pm to 6pm find out peak hr for for this freeway also find out the actual (design) flow rate for the actual traffic volume

time interval	traffic vol.
5:00-5:15	900
5:15-6:30	1000
5:30-5:45	1200
5:45-6:00 pm	850

★ very imppgt

q is peak hour volume (vehicle/hr) (6.8

q_{15} volume during the peak 15 minute of How. (unit vehicle/15 min) or (vehicle/$\frac{1}{4}^{th}$ hr)

For highway PHF = 0.8 - 0.95

Ans $PHF = \frac{1}{4} \times \frac{q}{q_{15}}$

q = peak hour volume (vehicle/hour)

= 900 + 1000 + 1200 + 850

= 3950 vehicle/hour.

q_{15} = 1200 vehicle/15 min.

$PHF = \frac{1}{4} \frac{3950}{1200} = 0.823$

Actual design flow rate = $\frac{q}{PHF}$.

$= \frac{3950}{0.823} = 4800$ vehicle/hr.

7.1

Speed Study Class 7 10 Aug

Speed study
- kyo kartehi
- Methods
  - ① eno/mirr
    - Mirrors kisanglepe set kretehi
    - enoscope distance frm observer
  - ② Other
    - a) electric
    - b) sptgn
    - c) elect mtr
    - d) time lapse
- Uses of spotspd
  - 1) plng trfc
  - 2) accs
  - 3) geo des
  - 4) trafcap

- Speed study is required to analyse highway geometrics.

- Methods

1. Mirror box method or enoscope

- Mirrors are set @ 45°
- Generally enoscope is kpt 50m from observer

2. other methods

a) photoelectric method

b) Radar gun / spot gun → Spot speed

c) electronic meter mthd

d) time lapse method

Uses of spot speed study

1) In planning & regulating traffic

2) In accident studies

3) In highway geometric design

4) to calculate traffic capacit